RELATIVE PSEUDO-COHERENCE IS LOCAL FOR A COVER BY PRINCIPAL OPENS OF THE ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 21668-21681 (canonical tag 067E) and the counted proof is the proof environment at source lines 21683-21712, which immediately follows it and is its own proof.

Editorial note. The printed proof begins a sentence with a lower case choose at source line 21686 and writes such that the elements map as described; the wording is copied exactly as printed and is not corrected.

Editorial note. The three references are all to earlier results of this chapter, namely the localization statement, the transfer of relative pseudo-coherence and the glueing statement; the kinds recorded are the type words appearing in the labels themselves. All three are navigation stubs with their roles recorded in the reasons.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- Relative pseudo-coherence of a complex is preserved by localizing the algebra at an element, which gives one of the two implications. invoked at proof lines 21683-21685. Disposes of the implication from the second to the first condition.
- The algebra can be presented by a polynomial ring in the elements and their complementary coefficients modulo the relation expressing that the elements generate the unit ideal. used at proof lines 21686-21695. Produces the polynomial presentation to which the transfers of relative pseudo-coherence are applied.
- The quotient presenting the algebra is pseudo-coherent as a module over the polynomial ring and remains so after inverting one of the elements, and relative pseudo-coherence transfers to a larger algebra under this hypothesis. used at proof lines 21695-21711. Moves the hypothesis from the localized algebras to the polynomial ring and back, giving the other implication.

Source reference. Original lemma 0679: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21499–21499. The reference is cited at proof lines 21683-21685 for the implication from relative pseudo-coherence to its localizations; the

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converse, which is the substance of the proof, is inside the counted span. The lemma itself is the subject of a separate card of this lane with disjoint coordinates.

Source reference. Original lemma 064Z: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16283–16283. The reference is applied twice, at proof lines 21699-21703 and at proof lines 21709-21711, to move relative pseudo-coherence between the algebra, its presentation and the polynomial ring; the presentation that makes it applicable is inside the counted span.

Source reference. Original lemma 066D: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16404–16404. The reference glues pseudo-coherence over the presentation from the localizations at the chosen elements, applied at proof lines 21704-21706; the localizations themselves are produced inside the counted span.

Lemma 0.2. *Let $R \to A$ be a finite type ring map. Let $K^\bullet$ be a complex of A-modules. Let $m \in \mathbf{Z}$. Let $f_1, \ldots, f_r \in A$ generate the unit ideal. The following are equivalent*

(1) *each $K^\bullet \otimes_A A_{f_i}$ is m-pseudo-coherent relative to R, and*
(2) *$K^\bullet$ is m-pseudo-coherent relative to R.*

The same equivalence holds for pseudo-coherence relative to R.

Proof. The implication (2) $\Rightarrow$ (1) is in Lemma 0679. Assume (1). Write $1 = \sum f_i g_i$ in A. Choose a surjection $R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r] \to A$. such that y_i maps to f_i and z_i maps to g_i. Then we see that there exists a surjection

$$P = R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r]/(\sum y_i z_i - 1) \longrightarrow A.$$

Note that P is pseudo-coherent as an $R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r]$-module and that $P[1/y_i]$ is pseudo-coherent as an $R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r, 1/y_i]$-module. Hence by Lemma 064Z we see that $K^\bullet \otimes_A A_{f_i}$ is an m-pseudo-coherent complex of $P[1/y_i]$-modules for each i. Thus by Lemma 066D we see that $K^\bullet$ is pseudo-coherent as a complex of P-modules, and Lemma 064Z shows that $K^\bullet$ is pseudo-coherent as a complex of $R[x_1, \ldots, x_n, y_1, \ldots, y_r, z_1, \ldots, z_r]$-modules. $\square$